

Table I
PTA organic impurities

Material	Amoco US	Amoco Geel	Mitsui	ICI	Cape Ind	Capco	Samsung	Temex	Interquisa	Jinshan
Year Tested			83, 86, 87	83, 85, 87	82, 85, 86	81	82, 84, 87, 88	81, 83	81, 82, 94	82
Liquid Chromatography	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm
4-CBA	10 - 20	10 20	5 16	3 - 6	11 – 17	16 23	16	11 - 12	9 - 16	2 – 9
p-Toluic Acid	100-140	110 140	100 - 150	60 130	0	125 145	260	25 - 135	To 140	110 - 150
Benzoic Acid	<2	<2	42 - 20	18	2 - 8	13	17	42		14
HM benzoic acid	10 - 30	6 - 15	1 - 27	0 - 6	10 - 20	17	25	2.2	9 - 14	4 - 7
ELC										
Trimellitic Acid	10 - 25	2 - 8	0.4 - 21	0.5 - 9	0.2 - 3	23	17	2.5	1 - 11	9 - 13
Isophthalic Acid	42	<2	<2	<2	<2 – 18	<2	15	<2	<2	5
TC benzophenone	0.1 - 2	<0.1	<0.1 - 0.11	<0.2 0.6	<0.1 - 1.3	<0.1	3	5	<2 – 3	<0.1 – 2
3,7 DC benzocoumarine	1 - 4	0	0.5 - 1.6	0.6 - 2.7	0 1	1.6	2.5		<0.1	0.5 – 1.5
TC biphenyl	15	5 - 7	1 - 11	8 - 12	1 - 4	12	20	9	11 – 16	10
2,6 CC Fluorenone	0.1 - 1	<0.1	<0.1 - 1.4	<0.1	0.1 - 1.1	0.1	4	<0.1	<0.1 – 0.3	<0.1
CC Biphenyl	60 - 90	50 - 110	40	60- 150	1.5 - 4		160	71	60 - 160	45- 70
2,6 CC Fluorene	15 - 30	20 - 40	13 - 23	4 - 36	2.2 - 6	21	25	2.9	17 – 21	10 - 20
Benzoic acid		0	3 - 13	3 - 16	0.4 - 2	7	5	0	2 – 4	3- -6
bis CP Methane	<0.1 3	<0.1 - 1.5	<0.1 - 0.8	<0.1 - 2	<0.1 - 1.5	1.1	1.4		<0.1 – 0.8	41
bis CP Ethane	1 5	2 4	0.4 - 7	1.5	0.4 - 0.6	3.8	6	4	4 – 6	5 – 12
CC Anthracene	0.2	0.2 - 0.9			<0.4	0.3	1.6		0.2	<0.2 – 0.6
Other					440 – 1000*					

HM = hydroxymethyl; CO = carboxy; TC = tricarboxy; DC = dicarboxy; CC = carboxydicarboxy; CP = carboxyphenyl;

LC = liquid chromatography; ELC = electrochromatography;

* Monomethyl terephthalate

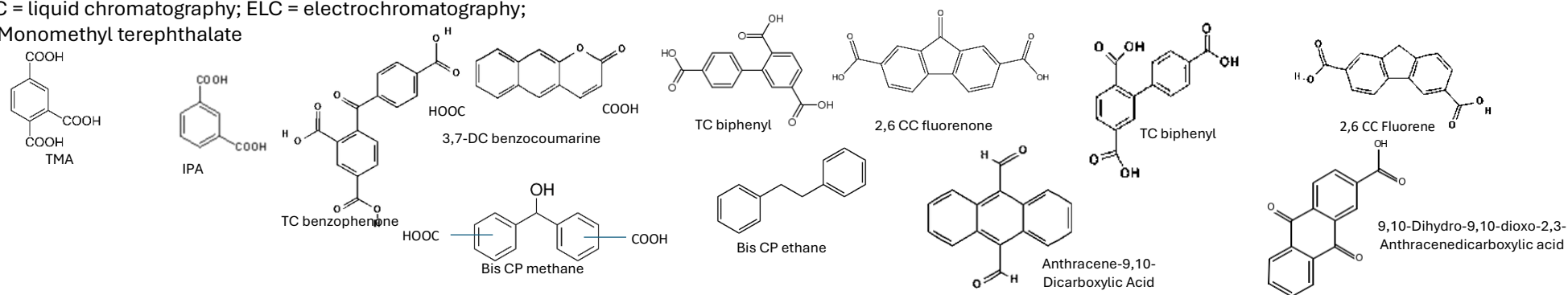


Table II
PTA inorganic impurities

Material	Amoco US	Amoco Geel	Mitsui	ICI	Cape Ind	Capco	Samsung	Temex	Interquisa	Jinshan
Year Tested			83, 86, 87	83, 85, 87	82, 85, 86	81	82, 84, 87, 88	81, 83	81, 82, 84	82
Ash ppm	1 - 3	2 - 4	1 - 4	2 - 5	1 - 2	2	3	2 - 9	6 - 9	6 - 8
metals										
Aluminum	<0.1	0.1	<0.1	<0.1	<0.1	0.2	<0.1	<0.1	<0.1	<0.1 - 0.2
Calcium	0.1 - 0.2	0.1 - 0.2	0.1 - 0.2	<0.2 0.9	<0.1	<0.1 - 0.2	0.1	<0.1	0.3	0.2
Cobalt	<0.1	<0.1	<0.1	<0.1 0.2	<0.05	0.05	<0.1 - 0.3	<0.2 - 0.3	0.1	<0.2
Chromium	<0.1	0.1	0.02	<0.1	<0.1 - 0.3	0.04	<0.1	<0.1	0.02	<0.2
Copper	0.01	<0.01	0.05	0.05	<0.04	0.1 - 0.2	<0.01	<0.1	<0.04	<0.01
Iron	0.2 - 0.4	0.1 - 0.4	0.02 - 0.2	0.1 - 0.3	<0.05 - 0.2	<0.02	<0.1-0.3	<0.01	0.1 - 0.2	<0.05
Potassium	<0.05	<0.05	0.01 - 0.06	<0.1 - 0.2		0.02	0.05	0.2	0.05	<0.05 - 0.3
Magnesium	<0.05	<0.05	4.05	<0.05	<0.05 - 0.2	0.96 - 1.0	<0.05	<0.2	<0.05	<0.05
Manganese	0.1 - 0.4	0.1 - 0.2	0.02,	0.1 - 0.7	<0.01	<0.01	0.2 - 0.8	<0.05	<0.09	0.01
Molybdenum	0.01	<0.01	<0.01	0.01	<0.1	<0.01	0.01	<0.01	0.3 - 0.4	0.1 - 0.3
Sodium	0.2 - 0.5	0.1 - 0.3	0.1 0.6	0.2 - 0.4	<0.1 - 0.6	0.7	0.2 - 0.3	0.2 - 0.4	<0.01	
Nickel	<0.1	<0.01	1	4.1 - 0.3	4.1	4.1	<0.1	<0.01	<0.02	<0.01
Titanium	<0.05	<0.05	0.05	<.05	<0.05	<0.05	<0.05	<0.05	<0.01	<0.05
Zinc	<0.01	0.01	<0.01 - 0.2	<0.01	<0.01	0.03	<.01	<0.01	<0.05	<0.08
Water	100 - 900	1000	60 1200	50 - 1200	450	570	300 4000	30 - 1200	360 - 1200	200 - 700
Delta Y	2 - 5	2 - 4	2 - 4	1.5 - 5	1.1 - 4.5	2.5 - 5	2 - 5	3		4 - 6

Table III
PTA Optical properties and particle size

Material	Amoco TA	Amoco	Mitsui TA	ICI	Cape Ind.	Capco	Samsung	Temex	Interquisa	Jinshan
Year Tested	U. S.	Geel TA-99	83, 86	81, 84	82, 84, 85	TA 84	QTA 81	S-QTA 81	MTA 87	87
L	98.6 – 99.0	98.8 – 99.0	98.7 – 99.1	98.6	98.9	98.8	98.7	98.7	98.8	98.3
a	-0.2 - -0.4	-0.5 - -0.7	-0.4 - -0.6	-0.4 -0.6	-0.4 -0.6	-0.4 -0.6	-0.4 -0.6	-0.4 -0.6	-0.4 -0.6	-0.2 -0.5
b	0.7 – 1.1	0.5 – 1.2	0.8 – 1.1	1.0 – 1.2	0.1 – 0.20-	0.9 – 1.5	1.0 – 1.7	1.0 – 1.5	1.0 – 1.6	0.5 – 1.2
Fluorescence Index	0.5 - 0.8	0.7 - 1.1	0.4 - 0.7	0.6 - 1.1	0 - 0.04	0.5 - 1.5	0.7 - 1.4	0.8 - 1.5	0.9 - 1.4	0.2 - 0.5
4-Ctr, ppm	0.3 - 0.5	0.7 - 1.2	0.1 - 0.2	0.1 - 0.7	0.0	0.81	0.2 - 1.0	1.08	0.34	0.1 - 0.2
340 nm, %	0.08 - 0.11	0.10 - 0.13	0.06 - 0.11	0.09 - 0.12	0.01 - 0.023	0.08 - 0.15	DAD - 0.15	0.040	0.09 - 0.16	0.08 - 0.11
400 nm, %	0.005 - 0.011	0.006- 0.011	0.001 - 0.010	0.011 - 0.012	0.001 - 0.003	0.006 - 0.015	0.003 - 0.019	0.004	0.007 - 0.012	0.008 - 0.011
Stability test @340 nm, %	69 78		78	67	97		55 - 70	96		66 - 96
Stability test @340 nm, %	88 94		93				80 - 90	99		91 - 98

Passing, mesh/micron	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large
70/210	98/96	95	95/85	88/95	88/95	95	93	96	98	95
100/149	88/83	76,	77/55	68/81	70/89	78	75/80	?9	81	42/51
140/105	68/59	52	51/53	40/60	64/80	57	52/60	50	60	30/35
200/74	48/37	32	32/19	22/38,	48/62	40	35/45	30	34	18/20
270/53	34/24	22	20/12	11/24	33/44	29	23/33	15	21	12/17
325/44	26/15	15	17/10	10/15	26/36	24	16/29	14	17	11/13
400/38	18/9	11	12/5	6/14	22/30	67	12/25	10	9	
Avg particle size	77/90	100	103/138	90/120	60/75	95	85/95	105	93	100 115

Table IV
TA and MTA inorganic impurities

Material	Amoco US	Amoco Geel	Mitsui TA	Capco TA	Samsung TA	Interquisa	Mitsubishi	Mitsubishi	Eastman
Year Tested	TA	TA	83, 86	81, 84	82, 84	TA 84	QTA 89, 89	S-QTA	MTA 89
Ash ppm	90 - 120	6	15 – 25	79	44	100	1 - 3		13
Metals, ppm	1	0.1	<0.2	<0.02	0.3	0.2	<0.1	<0.2 – 0.2	0.1
Aluminum	0.1 0.2	0.2	0.1 – 0.5	<0.1	0.2	0.06–0.4	<0.01	0.1	0.2
Calcium	5.2	1.9	7 – 11	8.0	6.3	27	02 – 04	0.45	2.9
Cobalt	<0.1 0.2	0.01	<0.1	0.2	0.1	0.1 – 0.2	<0.01	<0.01	<0.1
Chromium	0.02	0.02	0.02	<0.02	0.02	<0.02	<0.02	<0.02	<0.02
Copper	0.2 - 1.5	0.1	0.2 – 0.4	1.4	0.6	0.9 – 1.0	<0.01	<0.01	0.5
Iron	0 - 2	<0.1	0.1 – 0.4	<0.01	<0.01	<0.02	<0.01	<0.01	0.2
Potassium	<0.01	0.1		<0.01	<0.01	<0.1 – 0.15	<0.01		<0.01
Magnesium	30 - 10	3.0	35	53	37	50	0.2 – 0.5	0.2 – 0.6	0.1
Manganese	<0.01	<0.01	<0.01 – 0.15	<0.01	<0.01	0.01	<0.01	0.03	<0.01
Molybdenum	3 - 15	1.1	3	<0.1	2.9	2 – 9	<0.05	0.6 – 1.2	0.1
Sodium	0.1 - 0.8	<0.05	<0.05	0.2	0.05	0.2	<0.01	<0.05	0.1
Nickel	<0.1	<1	<0.1	<0.1	0.1	<0.01	0.3	<0.1	09
Titanium	<0.01	0.01	0.1 – 0.4	0.01	<0.01	<0.01		<0.01	.03
Zinc									
Bromine	30	9	50	25	24 – 30	52	42		
Water	50 100	100	100 – 200	100		200	250	100 – 260	150
Delta Y	3 - 45	9.6	1.3	13	4 - 9	3	2.3	4	3

Table V
TA and MTA organic impurities

Material	Amoco TA	Amoco	Mitsui TA	Capco TA	Samsung TA	Interquisa	Mitsubishi	Mitsubishi	Eastman	Eastman TA	DuPont TA
Year Tested	U. S.	Geel TA-99	83, 86	81, 84	82, 84, 85	TA 84	QTA 81	S-QTA 81	MTA 87	87	87
LC	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm	ppm
4-CBA	2500- 3000	2900	3200 - 3500	2700	1400 - 3300	1800 2700	250 - 300	60 -120	300	5100	250
p-Toluic Acid	200	240	450 - 700	350	200 - 550	300 - 500	20 - 60	5 - 50	60	440	7200
Benzoic Acid	150 - 300	12	200 - 300	180	100 - 500	250 - 400	30 - 40	30	80	450	30
HM benzoic acid	20 - 50		40 - 200	19	20 - 40	24 - 30			o	90	24
ELC											
Trimellitic Acid	400 - 600	76	3 - 760	540	490	190 - 280	50 - 100	20-70	140	250	7
Isophthalic Acid	100 - 500	920	150 - 1000	200	220	200 - 320	30 - 60	25	65	7200	231
TC benzophenone	30 - 50	9	35 - 45	50	40 - 60	10 - 40	24	3 - 15	4	30	se
3,7 DC benzocoumarine	5 - 20	1 - 3	27	4 - 1	6.3	1 - 16	5		9	24	31
TC biphenyl	80	40	70 - 100	110	140	50 - 130	20 - 45	10- 40	1s	16	161
2,6 DC Fluorenone	30 - 60	19	60 - 70	55	40 - 60	30 - 60	26	4	10	40	221
DC Biphenyl	70 - 150	58	715 - 110	140	130 - 140	50 - 180	40 - 60	30 -60	25	40	2
DC benzophenone	1040	3 - 7	14	27	16 - 35	17	2	'Cl		3	2!
DC Anthraquinone	410	3 - 5	57	11	10	11	1	-cl		2	1;
3,6 DC Fluorenone	10 - 20	2	11 - 3	16	12	13	2	42,		4	51
DC Stilbene	24	1	68	6	5	6	1	o			
Acetic Acid	300 - 700	40	900	1000	900	6800	700	750			1071

HM = hydroxymethyl; C = carboxy; TC = tricarboxy; DC = dicarboxy; CC = carboxydicarboxy; CP = carboxyphenyl;
LC = liquid chromatography; ELC = electrochromatography;

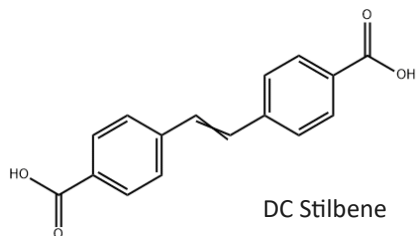


Table VI
TA Optical properties and particle size

Material	Amoco TA	Amoco	Mitsui TA	Capco TA	Samsung TA	Interquisa	Mitsubishi	Mitsubishi	Eastman	Eastman TA	DuPont TA
Year Tested	U. S.	Geel TA-99	83, 86	81, 84	82, 84, 85	TA 84	QTA 81	S-QTA 81	MTA 87	87	87
L	97	97.9	98.3 – 98.5	92.0	97.6 - 97.8	97.0 - 97.9	98.5 - " . O	98.8	98.8	97.8	97.3
a	-0.7 – 1.5	-0.8	-1. – 1.4	-1.6	-1.7 -1.9	-0.8 - 1.1	-0.4-0.6	-0.3	-0.8	-0.5	-1.7
b	4.7	3.8	4.8 – 5.2	4.7 – 8.4	6.5 7.8	5.0 5.8	2.0- 2.2	1 - 1.2	3.2	2.8	7.3
Fluorescence Index	0.4 – 0.8	0.2	0.5 – 0.6	0.7 – 1.4	0.5 1.0	0.5	0.1		0.2	0.3	0.2
4-Ctr, ppm	0.3 – 0.5	0.1	0.2	0.74	0.11 - 05	0.30	o. 1	40.1	0.05		1.2
340 nm, %	0.5 – 0.9	0.4	1.0 – 1.3	0.6 – 1.4	1.2 - 1.6	0.7 - 1.2	0.08 - 0.10	0.03 - 0.06	0.23	1.0	2.2
400 nm, %	0.07 – 0.12	0.05	0.15 – 0.18	0.06 – 0.15	0.2	0.09 - 0.13	0.01	4.01	0.04	0.1	0.3
Stability test @340 nm, %	25	21	41		3		50 - 60	25	40		
Stability test @340 nm, %	23 - 36	61	13		17 - 38		72 84		65	51	12

Passing, mesh/micron	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large	Small/large
70/210	95/91		89	65	58	76		97	100	92	
100/149	62/67	77	79	52	47	65	98		100		24
140/105	39/41	49	71	41	35	57	82	71	99	89	72
200/74	32/26	32	65	32	26	47	60	43		83	
270/53	27/15	15s	57	23	22	41	30	23	76		59
325/44	25/12	15	51	19	19	36	25	14	63	52	51
400/38	23/9	12	50	16	17	32	13	9	48	36	47
Avg particle size	133/130	107	39	145	165	85	65	50	40	42	42